MAT 1013: Discrete Mathematics

Session 1: Propositions and Logical Language

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Learning Outcomes

By the end of this session, students should be able to:

1. **Identify** whether a statement is a proposition
2. **Distinguish** propositions from non-propositions
3. **Understand** and **use** basic logical connectives
4. **Translate** simple English statements into symbolic form
5. **Recognize** ambiguity in everyday language

Part 1: Statements/Propositions

Of the following statements:

- Which are "mathematical"?
- Which can be classified as true or false?
- Which are problematic?

1. $2 + 3 = 5$
2. It is raining
3. $x + 2 = 5$
4. Close the door
5. This statement is false

Notes

- Ability to determine a "Truth Value" is the key criterion.
- Truth values are : **True** (T) and **False** (F)
- Some statements have variables and its truth value depends on the values of variables
- Commands and questions are not propositions
- Threre are "**Paradoxical** statements"

Key Definition

A **proposition** is a declarative statement that is either **true** or **false**, but not both.

Types of Statements

Type	Example	Reason
Proposition	$2+3=5$ $2+3=6$	Has truth value
Open sentence	$x>3$ $x^2+1=0$	Depends on variable
Commands Questions	Close the door What time is it?	No truth value No truth value

Part 2: Logical Connectives

Logical Connectives in Everyday Language

Logical connectives are used to combine statements and express relationships between ideas. Below are common connectives with examples from everyday language and mathematics.

Note: In the following "**Meaning**" referrers to under what conditions the compound statement is interpreted to be **True** (i.e., *Semantic*). The "**Form**" is the *Syntax*, the symbolic representation.

Conjunction (AND)

Symbol: $\wedge$ **Form:** $P \wedge Q$ **Meaning:** Both P and Q are true

Examples

Discuss the Truth Values:

- It is raining **and** it is cold.
- 2 is even **and** 3 is odd.
- Kamalani passed the exam **and** submitted all assignments.
- 1 is greater than 2 **and** 2 is prime.

Disjunction (OR)

Symbol: $\vee$ **Form:** $P \vee Q$ **Meaning:** *At least* one of P or Q is true

Note: *In logic, "or" is usually inclusive unless stated otherwise.*

Examples

Discuss the Truth Values:

- I study mathematics **or** physics.
- You can pay by cash **or** card.
- Ten is even **or** divisible by 3.

Negation (NOT)

Symbol: $\neg/\sim$ **Form:** $\neg P$ **Meaning:** not P is true (P is false)

Examples

Discuss the Truth Values:

- It is **not** raining.
- Nimal did **not** submit the assignment.
- 5 is **not** an even number.
- It is **not** the case that 2 is **not** prime.

Implication (IF...THEN)

Symbol: $\rightarrow/\Rightarrow$ **Form:** $P \rightarrow Q$ **Meaning:** If P is true, then Q is true

Examples

Discuss the Truth Values:

- **If** it rains, **then** the ground gets wet.
- **If** a number is divisible by 4, **then** it is even.
- **If** you get more than 80 percent for all the assignments, **then** you will pass the course with an A.

Note: *In classical logic, the "implication" IF P THEN Q (written symbolically $P \Rightarrow Q$) is True unless the antecedent P is True and the consequent Q is False. In particular if the antecedent is False, the implication is True irrespective to the truth value of the consequent. Therefore, this implication is often referred to as "Material implication".*

Examples

Discuss the Truth Values:

- **If** $2 = 3$, **then** 100 is the largest number.
- **If** there is a real solution to $x^2 + 1 = 0$, **then** then every quadratic equation with real constants has two solutions.
- **If** $x < 1$, **then** $x^2 < 1$

Notes: **Implication** does not mean **causation**.

Biconditional (IF AND ONLY IF)

Symbol: $\leftrightarrow / \Leftrightarrow$ **Form:** $P \leftrightarrow Q$ **Meaning:** P is true exactly when Q is true.

Examples

Discuss the Truth Values:

- A number is even **if and only if** it is divisible by 2.
- You can enter the room **if and only if** you have a valid ID.
- A shape is a square **if and only if** it has four equal sides and four right angles.

Activity : Biconditional

- Can you write the biconditional in terms of the other logical connectives?

Combining Connectives

Examples

Convert to the symbolic form. Note: You may want to use brackets to avoid different interpretations.

- If it is raining **and** cold, **then** I will stay home.
- A number is even **and not** divisible by 4.
- If a student studies **or** attends extra classes, **then** they will improve their marks.

Activity : XOR

Exclusive OR (XOR): "P XOR Q" is true *exactly when one* of P or Q is true (*but not both*).

Examples

- You can choose tea **or** coffee, but not both.
- The switch is either on **or** off (but not both).

- Can you write XOR as a combination the other logical connectives?

Logical Connectives Summary

Name	Symbol	Meaning
Negation	$\neg p$	Not p
Conjunction	$p \wedge q$	p and q
Disjunction	$p \vee q$	p or q (inclusive)
Implication	$p \to q$	If p, then q
Biconditional	$p \leftrightarrow q$	p iff q

Part 3: Practice

Task A: Identify Propositions

Classify each:

1. All birds can fly
2. $5 < 3$
3. Study hard
4. $y = 2x + 1$

Task B: Translation

Let:

$$p : \text{“I study”}, \quad q : \text{“I pass”}$$

Express using symbols:

1. I study and I pass
2. If I study, then I pass
3. I pass only if I study
4. I do not study

Important

“I pass only if I study”
means
“If I pass, then I (should have) studied”

Part 4: Activity – Statement Surgery

Analyze the following statements and

- Rewrite them clearly (if necessary)
- Identify the basic statements (P, Q, etc.)
- Express symbolically
- Decide truth value or explain why a truth value cannot be given
- Identify assumptions if there is any on which your answers might depend on.

1. Students who study pass exams
2. Everyone likes mathematics
3. You will succeed if you work hard
4. If a number is even, then it is divisible by 2.
5. If a number is divisible by 2, then it is even.
6. A number is even and greater than 10.
7. x is odd or divisible by 3.
8. It is raining and it is windy.
9. I will get wet only if it rains.
10. I will get wet if it rains.
11. If 2 is prime and 3 is odd then 5=6.
12. 2 is prime and 5 is even or 6 is even
13. It is not true that 10 is both even and divisible by 3.
14. 10 is not even or not divisible by 2.
15. If a number is not even, then it is odd.

16. It is not raining or it is cold.
17. The student did not pass or did not submit the assignment.
18. If a number is divisible by 4, then it is divisible by 2.
19. If it rains, then the ground is wet.
20. If a shape is a square, then it is a rectangle.
21. If a number is greater than 5, then it is positive.
22. A number is even if and only if it is divisible by 2.
23. A shape is a square if and only if it has four equal sides.
24. You can enter the room if and only if you have an ID.
25. A number is divisible by 6 if and only if it is divisible by 2 and 3.
26. A student passes if and only if they score more than 50.
27. If a number is even and greater than 2, then it is composite.
28. If a student studies or attends classes, then they will pass.
29. A number is prime or even.
30. If it is raining and cold, then I will stay at home.
31. A number is divisible by 2 or 3, and greater than 10.

Part 5: Reflection and Exit

Discussion Questions

- What makes a statement precise?
- Why is ambiguity problematic in mathematics?

Common Misconceptions

Misconception	Correction
All sentences are propositions	Only truth-valued statements are
OR is exclusive	OR is inclusive
If–then implies cause	It is logical implication
Variables allowed in propositions	Not unless fixed

Independent Study

Practice

1. Write 5 propositions from real life
2. Write 3 non-propositions and explain why
3. Translate 5 statements into symbolic form

Investigation

Why is precise language important in mathematics and computer science?